

Giant Finger: Visuo-proprioceptive congruent virtual legs for flying actions in virtual reality

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ABSTRACT

An immersive experience in virtual reality occurs when visual sense accompanies congruent somatosensation. However, as users should not fall down and physical abilities vary among individuals, existing methods may exhibit limited or predefined motion flexibility. To address this issue, we present Giant Finger, a flying action method that mimics human gait using two big virtual fingers. We confirmed its ownership via proprioceptive drift and questionnaires, and found that it performed better than existing methods in tasks such as flying and kicking.

Index Terms: Human-centered computing—VR locomotion techniques—Finger Walking In Place

1 INTRODUCTION

Virtual reality (VR) enables users to experience realistic simulations of virtual scenarios that exist beyond the physical world. However, if VR reflects real world physical abilities as they are, it can be difficult to fully enjoy movements that exceed human physical capabilities, such as flight or performing flying kicks, in VR.

Recent studies have proposed methods to enable actions in VR that are either difficult to perform in the actual world. They provide a highly realistic visual-somatosensory impedance for flight using full-body motions, such as Body-leaning method [4]. Some studies use bodily motions for realistic actions, but enable movements beyond human performance (e.g., the height users can jump [3] or the angle users can stretch while kicking [2]). However these approaches are still bound by human body capabilities and the type of actions or flexibility of motions is predefined [2, 3] or restricted [4].

Our new method of supporting flying action uses enlarged fingers that replace the legs. Because it does not require movement of the entire body, the method enables various actions. We believe that it can generate the sensation of enlarged fingers and can be incorporated into VR scenarios without harming the user's experience.

2 GIANT FINGER METHOD

We used Hi5 VR gloves to track finger movements and translate them into VR locomotion by scaling the relative position of each finger joint by 10 times to create Giant Finger (Fig. 1 (a)). In contrast to previous methods like Finger Walking In Place (FWIP), which required users to keep their fingers in contact with a touchpad, we designed Giant Finger to allow for more flexibility by using finger movements in the air. Fig. 1 (a) shows how the motions of the index and middle fingers are translated into walking motions. The finger with the lowest fingertip position serves as the pivot finger, and the distance traveled by its tip is reflected in the body's movement.

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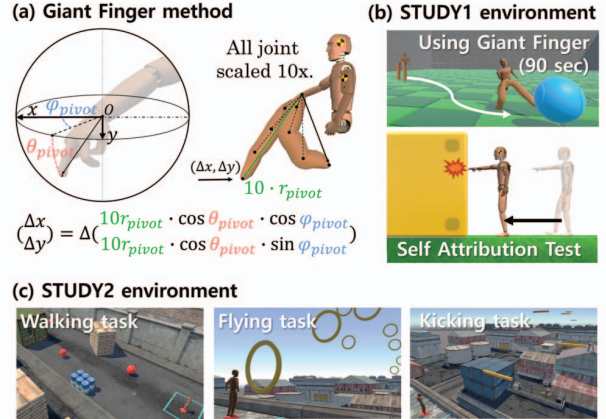


Figure 1: (a) illustration of the Giant Finger method, (b) and (c) VR environment used in Study 1 and 2.

Distance from wall (cm, $M \pm SD$)		t	p
Baseline condition (Before)	Giant Finger (After)		
59.2±15.7	69.5±18.7	3.10	0.003**
Questionnaire (ranging from -3 to 3)		t	p
Q1. I felt like I was walking on my big finger.	1.44±1.20	5.73	<.001***
Q2. I felt like my whole body was being moved by a large finger.	1.48±1.24	5.72	<.001***
Q3. I felt like my big fingers were mine.	1.35±1.50	4.32	<.001***
Q4. I felt like my real fingertips had moved to larger fingertip positions.	1.22±1.28	4.57	<.001***
Q5. I felt like my fingers located somewhere between my real fingers and my big ones.	0.30±1.55	0.94	0.178
Q6. I felt like my fingers were getting bigger.	1.04±1.55	3.22	0.002**
Q7. I felt like my big fingertips were moving to the actual fingertip position.	0.35±1.75	0.95	0.175

Figure 2: Results of proprioceptive drift and ownership questionnaire.

3 STUDY 1: VIRTUAL OWNERSHIP VERIFICATION

User study 1 verifies the ownership that users can have with enlarged virtual fingers using proprioceptive drift and an ownership illusion questionnaire. These validation approaches are commonly used as in a previous Rubber Hand Illusion (RHI) study [1].

3.1 Study Procedure

Participants were instructed to use Giant Finger to kick a ball for 90 seconds and to raise their arm in front while forming a “V” with their fingers (Fig. 1 (b)). They were then asked to walk towards a wall, stop when they believed their fingertip would touch it, and repeat three times with their body invisible in VR. Therefore, participants would stop at a greater distance from the wall if they perceived their fingers to be longer. The stopping distance was also measured before experiencing the Giant Finger method to construct the baseline condition. Then, participants filled out an ownership illusion questionnaire [1] using a 7-point Likert scale ranging from -3 to 3.

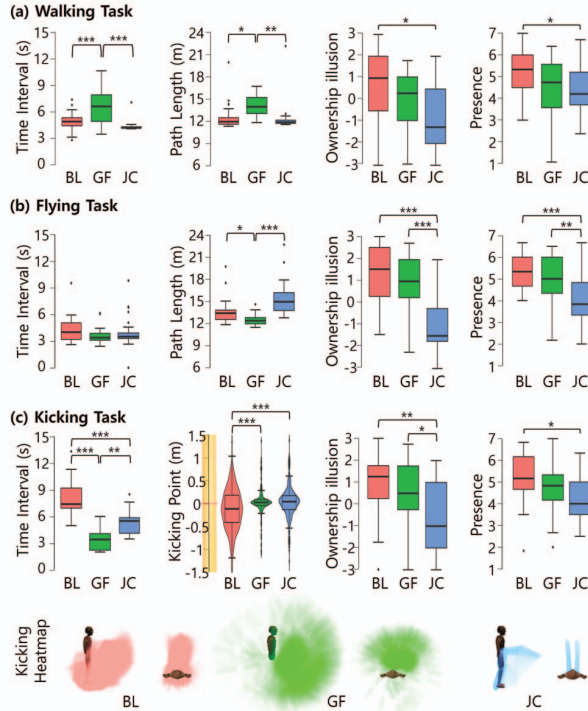


Figure 3: (a)Walking, (b)Flying, and (c)Kicking task. (BL:Body-leaning, GF:Giant Finger, JC: Joystick controller) (* $p < .05$, ** $p < .01$, *** $p < .001$)

3.2 Results and Discussion

We recruited 23 participants (18 males and 5 females) aged 21.3 ± 2.0 years and ensuring normality with skewness and kurtosis tests. The perceived location of the fingertips when participants raised their arms significantly increased after experiencing Giant Finger compared to the baseline according to the results of the one tailed ($> \text{baseline}$) paired samples t-test (Fig. 2). Subjective reports, as measured by an ownership illusion questionnaire, also supported this ownership. Participants reported their ownership for items Q1-4, and Q6 according to the results of the one tailed (> 0) one-sample t-test (Fig. 2). The RHI study [1] found that syncing real hand sensation with rubber hand stimulus creates ownership. Similarly, the Giant Finger creates a sense of ownership for the big finger by synchronizing real finger movement with enlarged virtual fingers.

4 STUDY2: UX WITH VARIOUS ACTION APPS

User study 2 compared our system to existing techniques for supporting flying actions in VR such as Body-leaning techniques and the Joystick-based method during walking, flying, and kicking tasks.

4.1 Set-ups for Flying Action Techniques

We used the Body-leaning and Joystick-based methods from Perusquia-Hernandez et al. [4]. The Body-leaning method uses body leaning for horizontal and head movement for vertical navigation, and VR trackers were placed on the feet for the kicking. The Joystick-based method uses an X-box controller for navigation and buttons for flying (Y and A buttons) and kicking (X and B buttons). Users can walk with Giant Finger by wriggling their fingers and can fly by varying the angle of the back of their hand, which is measured by the horizontal deviation between the fingertip and the wrist.

4.2 Study Procedure

Each participant completed the three tasks (Fig. 1 (c)) using three methods (Body-leaning, Giant Finger, Joystick controller) in a counter-balanced order. The walking task was a path with 10 check points. The flying task was a course with 10 rings at various heights.

The kicking task was a course with 10 bars in the air. All tasks were instructed to complete as quickly as possible. As the tasks progressed, the time and path length through the point or ring, the time to remove the bar, kicking accuracy and heatmap were collected. Then, participants filled out body ownership [1] and Slater, Usoh, Steed presence questionnaire using a 7-point Likert scale.

4.3 Results and Discussion

We recruited 25 participants (21 male and 4 female) aged 21.6 ± 2.0 years and used repeated measures ANOVA with Bonferroni test after checking for normality with skewness and kurtosis tests.

Walking Task As shown in Fig. 3 (a), GF took significantly longer to move between points than other methods. In addition, the length of the trajectory moved by GF was significantly longer than other methods. However, body ownership and presence did not show significant differences. GF may not be ideal for applications that value locomotion efficiency, but it provides similar body ownership and presence with minimal bodily actions while walking in VR.

Flying Task As shown in Fig. 3 (b), GF had significantly shorter trajectories than other methods. Our method also induced significantly higher ownership and presence than those of JC. In subjective ratings, the difference between GF and BL was insignificant and thus negligible. The results reveal that GF facilitates more efficiently for flight actions than other methods and has similar ownership and presence to BL, despite using only a portion of the body.

Kicking Task As shown in Fig. 3 (c), using GF, the participants explored considerably more available action spaces. In contrast, the BL permitted users to kick within the range of their physical capabilities, whereas JC operated within a predetermined range of motion. GF took significantly less time to complete kicking tasks than other methods. In addition, GF supported significantly more accurate targeting compared to BL. There was no significant difference between GF and JC in performance measures, but participants felt a greater sense of ownership with GF than with JC. The ownership of GF was comparable to that of BL. Our method allows efficient kicking interactions without compromising ownership or presence.

5 CONCLUSION AND FUTURE WORKS

We proposed Giant Finger, a method that outperforms existing ones in flying and kicking tasks in VR while maintaining comparable presence. It allows for more flexible flying actions using fingers, making physical ability less of a factor in performance. It is anticipated to have various applications, including supplementing physical abilities and actions beyond human capabilities. Since the current method necessitates constant usage of one hand, potential future work may involve the incorporation of manipulation tasks requiring a change in the mode of hand position and to permit different applications.

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